

Biogenic methane formation in marine and freshwater environments: CO₂ reduction vs. acetate fermentation—Isotope evidence

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Abstract—Two primary methanogenic pathways can be distinguished using the carbon and hydrogen stable isotope composition of the methane as a function of the coexisting carbon dioxide and formation water precursors. Although both pathways may occur in both marine and freshwater sediments, CO₂ reduction is dominant in the sulphate-free zone of the former, while acetate fermentation is the major pathway in freshwater sediments. Methane in marine sediments can be defined isotopically by $\delta^{13}\text{C}$ –110 to –60‰, and δD –250 to –170‰. In contrast, methane from freshwater sediments ranges from $\delta^{13}\text{C}$ –65 to –50‰ and δD –400 to –250‰. Carbon isotope fractionations ($\alpha_{\text{C-CO}_2\text{-CH}_4}$) are generally between 1.05 and 1.09 for marine sediments, while lower in freshwater sediments (1.04 to 1.06). The relationship of the methane to the formation water indicates the source of the hydrogen for CO₂ reduction to be the water directly with an associated hydrogen fractionation of $-180 \pm 20\%$.

The CH₄-H₂O hydrogen fractionation is larger for acetate fermentation due to the transfer of the methyl group during methanogenesis which is depleted in deuterium and accounts for 3/4 of the hydrogen in the methane. A model is presented showing that the fourth hydrogen *via* acetate fermentation may ultimately come from the formation water but is isotopically fractionated. Combination of the carbon and hydrogen isotope fractionations (α_{C} , α_{D}) from CH₄ with CO₂ and H₂O respectively, can clearly delineate the CO₂ reduction and acetate fermentation environments. Defining the character of the methanogenic types with carbon and hydrogen isotopes not only provides information about the environment of formation, it is also most useful in distinguishing biogenic from thermogenic methane gases.

INTRODUCTION

WITH THE RECOGNITION that over 20% of the world's natural gas accumulations are of a biogenic origin (RICE and CLAYPOOL, 1981), there is continued emphasis on understanding bacterial methane generation processes, particularly as an aid to hydrocarbon exploration.

Biogenic methane occurrences are a ubiquitous feature of recent anoxic sediments, and are well documented both in freshwater environments, such as lakes and swamps, and in marine environments, *e.g.* estuaries and shelf regions. Although early diagenetic formation of higher weight hydrocarbons including ethane and propane is possible, accumulations of biogenic natural gas are predominantly methane. In general, the total contribution of the higher homologs remains less than 1% of the methane. Unsaturated hydrocarbons also generated biogenically are not stable over geologic time and similarly constitute at most only a minor proportion of the total hydrocarbons in place. Thermal alteration of organic matter, which is responsible for much of the natural gas in oil-associated or certain gas reservoirs, can in the majority of cases be distinguished by the molecular and isotope composition of the gas from methane of bacterial origin. Transition and mixed gases certainly complicate the picture, and even the different origins of the organic matter, or subsequent alteration mechanisms of biogenic methane can produce a wide range of isotopic compositions. The purpose of this paper is, therefore, to discuss the different occurrences of biogenic methane in nature,

to examine the possible formation pathways and to demonstrate how these variations can be distinguished isotopically. Finally, these biogenic gases will be compared with thermogenic hydrocarbon gases in an overall natural gas isotope framework.

Occurrence of biogenic methane in sediments

Although methane is common in anoxic aqueous environments, significantly different distributions are observed between freshwater and marine sediments. Methane, among the most abundant gases generated in freshwater lakes and swamps, often increases rapidly in concentration reaching saturation levels within several centimeters sediment depth (*e.g.* REEBURGH and HEGGIE, 1977 or BARBER, 1974; CAPPENBERG, 1974; BOSSARD, 1981). Interstitial waters saturated with respect to methane can lead to bubble formation which is commonly observed in lakes and swamps as ebullition (*e.g.* ROSSOLIMO, 1935; STRAYER and TIEDJE, 1978).

In some freshwater environments, the bacterial production of methane is limited as found in Ace Lake (REEBURGH and HEGGIE, 1974) or Lake Mendota (WINFREY and ZEIKUS, 1977) and methane concentrations do not increase to saturation possibly due to substrate depletion or inhibition of methanogenesis. The methane concentration sometimes tends to stabilize or even decrease at greater depth. This phenomenon is either a consequence of methane consumption, or may represent non-steady state diagenesis in these sediments.

Biogenic methane in marine and brackish environments, such as salt marshes, estuaries, shelf and deep-sea sediments, have distribution constraints different

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to those in the freshwater milieu. With few exceptions, significant methane accumulations are measured at the sediment intervals where dissolved sulphate has been depleted to concentrations of less than 1.0 mM (NISENBAUM *et al.*, 1972; MARTENS and BERNER, 1977; WHITICAR, 1978; REEBURGH, 1980; LEIN *et al.*, 1981; and others). In marine sediments having higher accumulation rates and thus higher organic carbon contents (MÜLLER and SUESS, 1979), microbial dissimilation of organic matter rapidly utilizes the free oxygen, nitrate and manganese and iron oxides (FROELICH *et al.*, 1979), leaving dissolved sulphate the next available electron acceptor to be reduced. In such organic rich sediments, the dissolved sulphate concentration can also decrease quickly with depth and is frequently exhausted at sediment depths shallower than 2 meters, below which corresponds to the onset of methane accumulation.

Analogous to the freshwater sediments, methanogenesis in marine sediments can steadily increase the methane concentration to saturation and give rise in shallow water sediments to bubble formation and the creation of acoustically turbid sediment horizons with free gas (SCHUBEL, 1974; WHITICAR, 1982). Methane ebullition from shallow marine sediments has been reported (WHELAN, 1974; MARTENS, 1976; KING and WIEBE, 1978) but is generally prevented in deep-water environments by the higher hydrostatic pressure, (*e.g.* Santa Barbara Basin, BARNES and GOLDBERG, 1976). Gas hydrate formation in colder and/or deeper-water sediments can restrict methane ebullition (CLAYPOOL and KAPLAN, 1974), and lead to vast accumulations of natural gases, *e.g.* in Siberia and northern Canada (MAKOGON *et al.*, 1972).

Methanogenesis is limited in some marine sediments such that methane does not reach saturation levels. The explanation is similar to the possibilities given above for freshwater sediments.

Anaerobic methane oxidation

Bacterial oxidation of methane can result in the consumption of large amounts of the hydrocarbon and is associated with an isotope effect which can lead to large carbon and hydrogen isotope fractionations. In freshwater environments, the majority of methane consumption occurs in the oxic zones of sediment or water columns (RUDD *et al.*, 1974; BOSSARD, 1981). Based on geochemical evidence, methane consumption in marine sediments is primarily at the base of the zone of sulphate reduction and that methane from greater sediment depth is lost upon entering this zone (REEBURGH, 1976; MARTENS, 1982; WHITICAR, 1982; DEVOL, 1983). Although studies by DAVIS and YARBROUGH (1966), PANGANIBAN *et al.* (1979) and KO-SIUR and WARFORD (1979) have shown anaerobic oxidation to be possible using sulphate as oxidant, this anaerobic methane oxidation remains a poorly understood phenomenon in both freshwater and marine anoxic sediments. The subject of isotope fractionation

related to methane oxidation is dealt with in a separate paper (WHITICAR and FABER, 1985). Care was exercised in the compilation of the data base used for this paper to avoid samples which have sustained methane oxidation effects.

Biogenic methane formation pathways

Biogenic methane is an ultimate dissimilation product of microbially mediated reactions of organic molecules which accumulate in anoxic sediments. Two primary metabolic pathways by the biological group Methanogens (see BALCH *et al.*, 1979) are generally recognized for methanogenesis namely: fermentation of acetate and reduction of carbon dioxide (Fig. 1). Recent studies by HIPPE *et al.* (1979), KING *et al.* (1983), WINFREY and WARD (1983), OREMLAND (1985), ZEIKUS *et al.* (1985) have indicated that other substrates such as methylated amines, methanol and carbon monoxide can also be important for methanogenesis, in addition to ethanol, benzoate, and formate reported earlier (*e.g.* STADTMAN and BARKER, 1949; NOTTINGHAM and HUNGATE, 1969; SMITH and MAH, 1978). For simplicity in this paper, acetate fermentation refers collectively to methanogenesis involving the transfer of a methyl group from any substrate.

Acetate fermentation, was described as early as 1887 by HOPPE-SEYLER and later by BARKER (1936), BUSWELL and SOLO (1948), and STADTMAN and BARKER (1949), using ^{14}C labelling experiments represented by the general equation:



Here the methane is derived from the methyl group of acetate (PINE and BARKER, 1956; SMITH and MAH, 1980). Alternatively, acetate may be oxidized to CO_2 and H_2O , and subsequently the CO_2 metabolically reduced to CH_4 with hydrogen as electron source. Acetate

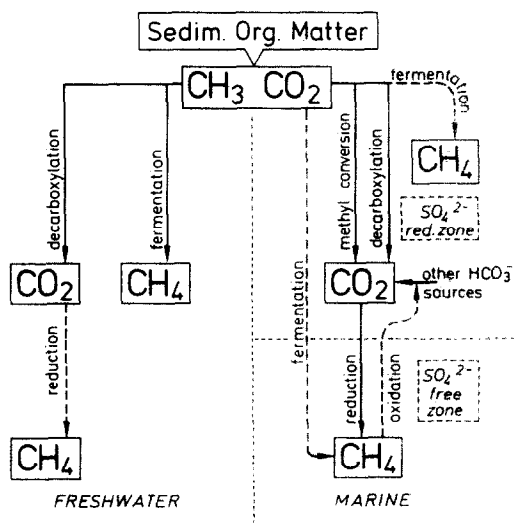
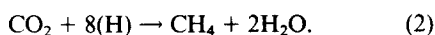


FIG. 1. Flow diagram comparing methanogenesis pathway by acetate fermentation and CO_2 reduction in freshwater and marine sediments.

fermentation is regarded by many investigators as the major source of methane (about 70%) in freshwater environments, such as sediments (KOYAMA, 1964; TAKAI, 1970; BELYAEV *et al.*, 1975; WINFREY *et al.*, 1977; CAPPENBERG and JONGEJAN, 1978); sewage sludges (JERIS and MCCARTY, 1965; SMITH and MAH, 1966; MOUNTFORT and ASHER, 1978) and in numerous culture studies (CAPPENBERG and PRINS, 1974; WEIMER and ZEIKUS, 1978; ZINDER and MAH, 1979; SMITH and MAH, 1978, 1980). As mentioned before there is evidence, however, that other substrates may be important, and that CO₂ reduction can be the primary methane source in certain low sulphate freshwater environments such as hot-spring algal mats (SANDBECK and WARD, 1981) and in bovine rumen (OPPERMANN *et al.*, 1961).

Schematically depicted in Fig. 1, the methyl position of acetate in anoxic freshwater sediments is fermented to methane. The carbon dioxide released by this acetate dissimilation may also be reduced to methane, though perhaps on a diminished scale. This CO₂ reduction could continue to operate even after acetate is exhausted and the acetate fermentation pathway ceases.

The mode of methane generation is somewhat different in the saline environments such as salt marshes and marine sediments (Fig. 1). Based on culture studies, the reduction of CO₂ by hydrogen is considered here to be the dominant pathway for bacterial methane formation (BELYAEV *et al.*, 1977; OREMLAND and TAYLOR, 1978):



Stimulation of methanogenesis by hydrogen addition has been demonstrated by MYLROIE and HUNGATE (1955), OREMLAND and TAYLOR (1978), WINFREY and ZEIKUS (1977) and ABRAM and NEDWELL (1978a,b). This hydrogen addition was shown by WINFREY *et al.* (1977) to increase methane production *via* CO₂ reduction but had no effect on the fermentation rate.

As noted earlier, significant methane accumulations in marine sediments are usually not observed in the presence of oxygen or nitrate, nor where sulphate contents are higher than 1.0 mM. This spatial partitioning of sulphate and methane lead to the earlier postulation of a noncommensurate behaviour between sulphate reducing and methanogenic bacteria. In contrast to the early results of BARBER (1974), CAPPENBERG (1974) and WINFREY and ZEIKUS (1977) showing methanogenesis inhibition by sulphate, mutual coexistence has been demonstrated by LAWRENCE *et al.* (1964), LAWRENCE and MCCARTY (1965), POLCIN *et al.* (1981) and LOVLEY *et al.* (1982). Methanogenesis in some freshwater environments is reported by CAPPENBERG (1975) to be inhibited by the relatively low sulphide activities. However, INGVOSEN and BROCK (1982) have shown that methane and sulphide can coexist, and MAH *et al.* (1977) reported that sulphide in certain instances can actually stimulate methanogenesis.

In marine sediments the often higher sulphide concentration does not appear to inhibit methanogenesis

(NIKAIDO, 1977; OREMLAND and TAYLOR, 1975). Although methanogens and sulphate reducing bacteria could coexist within the sulphate reducing zone, there is strong evidence that methanogenesis by fermentation is severely limited by substrate competition (*e.g.* for acetate, WINFREY and ZEIKUS, 1977; SCHÖNHEIT *et al.*, 1982) and that successful competition for hydrogen by the sulphate reducing bacteria over the methanogens restricts the methanogenesis by CO₂ reduction pathway (WINFREY and ZEIKUS, 1977; ABRAM and NEDWELL, 1978a,b; LOVLEY *et al.*, 1982).

Another important metabolic difference between marine and freshwater sediments is that in the presence of higher amounts of sulphate, the methyl position of acetate is oxidized to CO₂ rather than being fermented to methane. This acetate conversion to CO₂, observed by WINFREY and ZEIKUS (1977), MOUNTFORT and ASHER (1978), MOUNTFORT *et al.* (1980), KING and WEIBE (1978), WINFREY and WARD (1983), and NEDWELL (1982) is reportedly mediated by sulphate reducing bacteria capable of oxidizing acetate.

This conversion of methyl to CO₂ further enlarges the bicarbonate pool available to CO₂ reducing bacteria, which become active following the exhaustion of sulphate.

Although both methane production pathways can operate in freshwater and marine sediments, the relative proportions of each pathway are variable and controlled by the availability of substrates and hydrogen to the methanogens, which in turn can be limited by competition from sulphate reducing bacteria.

These pathways of biogenic methane formation can be further substantiated by the use of carbon and hydrogen isotopes. As discussed next, the combination of isotopic measurements from freshwater and marine sediments on methane with the coexisting CO₂ and H₂O produce distinctive fractionations which can delineate the various environments and pathways.

HYDROGEN AND CARBON ISOTOPE VARIATIONS

The data base

To test the general applicability of our hypotheses, a data base was created using the available hydrogen and carbon isotopic data on methane, carbon dioxide and the formation water from freshwater and marine situations. The data, a compilation of "in house" and published data from other workers, are grouped in Table 1 into various freshwater and marine (saline) environments, and experimental laboratory studies.

The isotope data selected were only those in sediments clearly in the primary methanogenesis region and judged to be free from major alteration effects such as oxidation. Although this compilation is not exhaustive, the purpose is to obtain representative values for the different natural environments.

For an analytical description of the isotope methods employed, refer to FABER and STAHL (1983) and to the source publications in Table 1. The carbon and hydrogen isotopic

Table 1a: Biogenic methane data base

SAMPLE	LOCATION	TEMP	$\delta^{13}\text{C}_1$	$\delta^{13}\text{CO}_2$	a_{C}	$\delta\text{D}_{\text{C}_1}$	$\delta\text{D}_{\text{H}_2\text{O}}$	a_{D}	REFERENCE
NAME		$^{\circ}\text{C}$	$^{\circ}/_{\text{oo}}$	$^{\circ}/_{\text{oo}}$		$^{\circ}/_{\text{oo}}$	$^{\circ}/_{\text{oo}}$		
FRESHWATER SWAMP									
VOLO BOG		15.0	-58.5	.	.	-324	.	.	WOLTEMAE, 1982
VOLO BOG		15.0	-64.3	.	.	-296	.	.	
VOLO BOG		15.0	-63.1	.	.	-263	-52.4	1.287	
VOLO BOG		15.0	-58.4	.	.	-224	-52.4	1.223	
VOLO BOG		15.0	-69.0	.	.	-273	-43.5	1.316	
VOLO BOG		15.0	-69.0	.	.	-232	.	.	
ALTWARMB.MOOR		17.0	-58.4	.	.	-374	-25.0	1.558	
ALTWARMB.MOOR		17.0	-60.3	.	.	-340	-25.0	1.477	
ALTWARMB.MOOR		17.0	-62.2	.	.	-352	-25.0	1.506	
BRAKE		12.0	-68.0	.	.	-291	-25.0	1.377	
SIBERIA MARSH 1		.	-56.4	-16.2	1.043	.	.	.	VOYTOV et al., 1975
SIBERIA MARSH 2		.	-60.6	-20.6	1.043	.	.	.	
SIBERIA MARSH 3		.	-53.7	-16.2	1.040	.	.	.	
SIBERIA MARSH 4		.	-56.2	-17.2	1.041	.	.	.	
SIBERIA MARSH 5		.	-58.9	-11.3	1.051	.	.	.	
SIBERIA MARSH 6		.	-49.9	-15.4	1.036	.	.	.	
SIBERIA MARSH 7		.	-61.7	-18.0	1.047	.	.	.	
SOVIET BIOGENIC GASES max.		.	-90.0	.	.	-400	.	.	ALEKSEYEV, 1980
SOVIET BIOGENIC GASES min.		.	-60.0	.	.	-250	.	.	
KOSHELEWKA SOIL GAS, USSR		.	-72.0	-23.0	1.053	.	.	.	LEBEDEW et al., 1969
FRESHWATER LAKE SEDIMENTS									
QUASSAPAU		7.8	-68.0	-25.2	1.046	.	.	.	OANA and DEEVEY, 1960
LINSLEY POND(17/5)		7.1	-77.4	-20.6	1.062	.	.	.	
LINSLEY POND(4/6)		7.0	-77.8	-7.1	1.077	.	.	.	
LINSLEY POND(24/8)		7.4	-80.2	-6.	1.081	.	.	.	
LINSLEY POND(8/10)		7.4	-77.5	-7.2	1.076	.	.	.	
WONONSCOPONUC(22/7)		5.3	-59.7	-6.1	1.057	.	.	.	
WONONSCOPONUC(18/9)		5.7	-60.8	-14.7	1.049	.	.	.	
QUEECHY(27/9)		7.0	-57.2	-15.2	1.045	.	.	.	
WG1.1. Wuermsee,		19.0	-60.2	-3.0	1.061	-291	.	.	WOLTEMAE et al., 1984
WG1.2. N. Germany		19.0	-60.2	-3.2	1.061	-295	.	.	
WG1.3		19.0	-55.6	-4.5	1.054	-305	.	.	
WG1.4		19.0	-57.8	-7.9	1.053	-294	.	.	
WG2.1		18.8	-58.0	-8.8	1.052	-318	.	.	
WG2.2		18.8	-57.8	-9.3	1.051	-308	.	.	
WG2.3		18.8	-59.9	-9.7	1.053	-309	.	.	
WG2.4		18.8	-59.3	-7.1	1.056	-312	.	.	
WG4.1		19.0	-56.0	-11.4	1.047	-320	.	.	
WG4.2		19.0	-56.2	-15.2	1.044	-321	.	.	
WG4.3		19.0	-57.3	-17.1	1.043	-321	.	.	
WG4.4		19.0	-56.9	-9.1	1.051	-319	.	.	
WG6.2		19.0	-55.8	.	.	-324	.	.	
WG7.1		18.0	-53.8	-5.1	1.052	-321	.	.	
WG7.2		18.0	-54.9	-4.7	1.053	-322	.	.	
WG7.3		18.0	-54.0	-3.6	1.053	-323	.	.	
WG7.4		18.0	-54.7	-3.7	1.054	-325	.	.	
WG5.1		17.5	-57.5	-13.8	1.046	-329	-21.8	1.459	
WG5.3		17.5	-57.4	-17.9	1.042	-323	-21.7	1.446	
WG8.1		18.5	-53.4	-7.6	1.048	-332	.	.	
WG8.2		18.5	-53.9	-7.9	1.049	-334	.	.	
WG8.3		18.5	-52.1	-8.5	1.046	-338	.	.	
WG8.4		18.5	-52.4	-9.8	1.045	-331	.	.	
WG9.1		18.5	-60.4	-14.0	1.049	-328	-24.6	1.452	
WG9.2		18.5	-60.5	-12.4	1.051	-331	-27.3	1.454	
WG9.3		18.5	-57.8	-13.5	1.047	-333	.	.	
WG9.4		18.5	-58.0	-15.4	1.045	-332	.	.	
WG10.1		17.5	-59.4	-15.1	1.047	-316	.	.	
WG10.2		17.5	-57.7	-14.6	1.046	-311	.	.	
WG10.3		17.5	-60.0	.	.	-318	.	.	
WG14.1		16.0	-58.3	-21.1	1.040	-329	.	.	
WG14.2		16.0	-58.6	-9.8	1.052	-331	.	.	
WG14.3		16.0	-57.0	-13.5	1.046	-332	.	.	
WG14.4		16.0	-55.2	-14.4	1.043	-334	.	.	
WG16.1		17.5	-58.2	-11.5	1.050	-323	.	.	
WG16.2		17.5	-58.6	-12.5	1.049	-326	.	.	
WG16.3		17.5	-58.2	-10.9	1.050	-335	.	.	
WG16.4		17.5	-58.9	-13.0	1.049	-334	.	.	
WG18.1		18.0	-58.3	-10.3	1.051	-327	-20.0	1.457	
WG18.2		18.0	-58.7	-11.5	1.050	-329	-25.0	1.455	
WG18.3		18.0	-58.0	-12.0	1.049	-331	.	.	
WG18.4		18.0	-59.9	-13.0	1.050	-324	.	.	
WG19.4		19.0	-64.0	-15.3	1.052	-323	-28.1	1.438	
WG20.1		17.5	-58.9	-15.6	1.046	-313	.	.	
WG20.2		17.5	-59.5	-16.1	1.046	-316	.	.	
WG20.3		17.5	-58.9	-13.8	1.048	-318	.	.	
WG20.4		17.5	-59.0	-10.8	1.051	-319	.	.	
WG21.1		17.5	-58.3	-10.8	1.051	-326	.	.	
WG21.2		17.5	-58.6	-11.2	1.050	-327	.	.	
WG21.3		17.5	-58.9	-10.8	1.051	-326	.	.	
WG21.4		17.5	-60.0	-11.6	1.051	-322	.	.	
WG23.1		15.5	-59.2	-14.1	1.048	-313	.	.	
WG23.2		15.5	-56.7	-12.7	1.047	-314	.	.	

Table 1b: Biogenic methane data base

SAMPLE NAME	LOCATION	TEMP °C	$\delta^{13}\text{C}_1$ ‰	$\delta^{13}\text{CO}_2$ ‰	α_C	$\delta\text{D}_{\text{C}_1}$ ‰	$\delta\text{D}_{\text{H}_2\text{O}}$ ‰	α_D	REFERENCE
GLACIAL DRIFT GAS									
ILLINOIS, USA		10.0	-82.7	.	.	-245	-47.0	1.262	SCHOELL, 1980
ILLINOIS, USA		10.0	-91.0	.	.	-226	.	.	
ILLINOIS, USA		10.0	-72.8	.	.	-239	-40.0	1.261	
ILLINOIS, USA		10.0	-73.0	.	.	-234	-50.0	1.240	
ILLINOIS, USA		10.0	-76.0	.	.	-239	.	.	
ILLINOIS, USA		10.0	-75.7	.	.	-277	-85.0	1.266	
GLACIAL DRIFT GAS			-75.0	-7.9	1.073	.	.	.	COLEMAN, 1976
MARINE MARSH									
PHIL/10 MARGOT		20.0	-76.2	.	.	-207	.	.	SCHOELL, 1980
PHIL/31 MARGOT LALA		20.0	-76.2	.	.	-216	.	.	
GERM/BRAKE MARSH GAS		12.0	-68.8	.	.	-244	-60.0	1.243	
SHALLOW MARINE									
SAANICH INLET		9.0	-55.6	16.7	1.077	.	.	.	NISSENBAUM et al., 1972
15550-3 Baltic Sea		.	-84.6	.	.	-219	-25.4	1.248	
WHITICAR and WERNER, 1981									
15550-4		.	-84.5	.	.	-247	-27.0	1.293	
15550-5		.	-82.5	.	.	-247	-28.5	1.291	
15550-6		.	-80.8	.	.	-246	-27.7	1.291	
15550-7		.	-79.6	.	.	-243	-27.1	1.285	
15551-8		.	-77.9	.	.	-234	.	.	
15551-9		.	-81.4	.	.	-250	.	.	
15551-10		.	-82.0	.	.	-237	.	.	
15551-11		.	-84.3	.	.	-229	.	.	
BH2 14 Gulf of Mexico		.	-80.1	-12.8	1.073	.	.	.	WHELAN et al., 1978
BH2 35		.	-82.3	-.9	.	.	.	.	
BH2 50		.	-53.2	-5.0	.	.	.	.	
BH2 65		.	-65.6	-17.0	.	.	.	.	
GAS-024		.	-90.4	-23.0	1.074	.	.	.	DOOSE, 1980
GAS-024		.	-85.4	-17.0	1.075	.	.	.	
GAS-025		.	-85.7	-13.7	1.079	.	.	.	
GAS-025		.	-81.9	-10.9	1.077	.	.	.	
GAS-025		.	-81.8	-7.1	1.081	.	.	.	
GAS-029		.	-76.8	-26.3	1.055	.	.	.	
GAS-029		.	-92.8	-20.2	1.080	.	.	.	
SOG 5T		.	-89.4	-16.9	1.081	.	.	.	
DEEP MARINE									
Deep Sea Drill Project									
11-102-5 Blake-Bahama		.	-88.4	-32.1	1.062	.	.	.	CLAYPOOL et al., 1973 and
11-102-11 Ridge		.	-72.8	1.0	1.080	.	.	.	LYON, 1974 and
11-102-19		.	-77.1	-7.9	1.075	.	.	.	FRIEDMAN and HARDCASTLE, 1973
11-104-3		.	-76.3	-23.2	1.057	.	.	.	
11-104-7		.	-70.7	-14.6	1.060	.	.	.	
11-104-9		.	-70.1	-14.4	1.060	.	.	.	
11-106-B-2 L.Continental		.	-86.1	-23.2	1.069	.	.	.	
11-106-B-5 Rise		.	-73.7	-7.4	1.072	.	.	.	
14-144-A-6 Demerara Rise		.	-70.8	-20.2	1.054	.	.	.	
15-147-9 Cariaco Trench		.	-65.0	3.7	1.073	.	.	.	
15-147-12		.	-61.7	6.5	1.073	.	.	.	
15-147-18		.	-65.0	10.5	1.081	.	.	.	
15-147-C-7		.	-64.7	6.9	1.077	.	.	.	
15-147-B-6-2		.	-67.6	-3.1	1.069	-179	11.2	1.232	
15-147-B-7-4		.	-65.2	-4.2	1.065	-173	8.1	1.219	
15-147-10		.	-64.6	-2.3	1.067	-175	9.9	1.224	
15-147-B-9-4		.	-60.8	-1.6	1.063	.	9.9	.	
15-147-B-11-3		.	-65.8	.1	1.070	-179	8.2	1.228	
15-147-14-3		.	-59.6	-.7	1.063	-179	.	.	
15-147-C-2-2		.	-65.1	.2	1.070	-181	4.0	1.226	
15-147-C-4-4		.	-64.4	.3	1.069	-183	6.6	1.232	
15-147-C-7-3		.	-64.6	-.8	1.068	-176	6.3	1.221	
15-154-A-7 Columbian		.	-81.3	-23.8	1.063	.	.	.	
15-154-5 Abyssal Rise		.	-69.1	-22.4	1.050	.	.	.	
15-154-9		.	-71.9	-24.8	1.051	.	.	.	
18-174 Astoria Fan		.	-88.0	-23.0	1.071	.	.	.	CLAYPOOL and KAPLAN, 1974
18-174		.	-77.0	-10.0	1.073	.	.	.	
18-174		.	-78.0	-9.0	1.075	.	.	.	
18-174		.	-73.0	-2.0	1.077	.	.	.	
18-174		.	-70.0	-1.0	1.074	.	.	.	
18-174		.	-71.0	-7.0	1.069	.	.	.	
18-174		.	-75.0	-19.0	1.061	.	.	.	
18-180 Aleutian Trench		.	-70.0	-6.0	1.069	.	.	.	
18-180		.	-81.0	-2.0	1.086	.	.	.	
64-479-3-1 Gulf of		5.2	-73.1	-4.3	1.074	-191	-1.3	1.235	SCHOELL, 1982
64-479-6-4 California		9.4	-66.6	1.0	1.072	-187	.	.	
64-479-9-4		13.2	-64.6	1.8	1.071	-187	-1.0	1.230	
64-479-12-6		17.0	-63.6	3.4	1.072	-183	-1.3	1.223	
64-479-15-2		20.0	-62.4	3.2	1.070	-178	.	.	

Table 1c: Biogenic methane data base

SAMPLE	LOCATION	TEMP	$\delta^{13}\text{C}_1$	$\delta^{13}\text{CO}_2$	α_{C}	$\delta\text{D}_{\text{C}_1}$	$\delta\text{D}_{\text{H}_2\text{O}}$	α_{D}	REFERENCE
NAME		$^{\circ}\text{C}$	‰	‰		‰	‰		
64-479-21-5		27.5	-62.4	.5	1.067	-174	-1.6	1.210	
64-479-23-5		29.7	-54.8	1.5	1.060	-169	.	.	
64-479-27-5		34.6	-62.7	-2.5	1.064	-171	-2.0	1.205	
64-479-31-5		39.5	-62.7	-3.9	1.063	-172	.	.	
64-479-37-4		46.4	-61.2	-4.2	1.061	-170	-4.4	1.201	
64-479-43-2		53.2	-61.5	-6.7	1.058	-178	.	.	
64-479-47-3		58.4	-60.9	-10.1	1.054	-185	.	.	
64-481-P4-1		5.7	-70.6	-5.1	1.071	-191	.	.	
64-481A-4-2		12.9	-66.3	-7.3	1.063	-178	-3.8	1.213	
64-481A-6-3		15.6	-69.4	.	.	-184	-5.6	1.220	
64-481A-9-5		23.7	-52.5	-3.3	1.052	-186	-1.4	1.228	
64-481A-12-5		28.0	-52.5	.	.	-175	.	.	
64-481A-14-6		31.5	-57.5	.	.	-178	.	.	
64-481A-20-2		39.8	-60.2	-3.3	1.061	-173	.	.	
64-481A-26-6		49.8	-61.6	.	.	-172	-1.3	1.207	
64-481A-30-7		56.0	-60.4	.	.	-174	2.1	1.214	
76-533-A-1-6	Blake Outer Ridge	.	-81.3	-11.9	1.076	.	.	.	CLAYPOOL and THRELKELD, 1983
76-533-20-1		.	-81.3	-11.1	1.076	.	.	.	
76-533-22-1		.	-79.8	-8.7	1.077	.	.	.	
76-533-24-2		.	-78.0	.	.	.	.	.	
76-533-28-3		.	-75.9	.	.	.	.	.	
76-533-34-1		.	-73.5	-3.5	1.076	.	.	.	
76-533-40-2		.	-71.3	-2.0	1.075	.	.	.	
76-533-A-5-6		.	-71.3	-4.5	1.072	.	.	.	
76-533-A-11-4		.	-68.2	-2.6	1.070	.	.	.	
76-533-A-15-2		.	-68.1	-2.9	1.070	.	.	.	
76-533-A-18-4		.	-67.1	-6.3	1.065	.	.	.	
76-533-A-20-6		.	-68.1	-4.8	1.068	.	.	.	
76-533-A-24-5		.	-67.5	-4.3	1.068	.	.	.	
76-533-A-26-4		.	-67.2	-3.8	1.068	.	.	.	
76-533-A-29-1		.	-67.6	-2.9	1.069	.	.	.	
95-603D-1-3		.	-79.6	.	.	-205	.	.	WHITICAR and FABER, 1986
95-603D-1-4		.	-79.7	.	.	-261	.	.	
95-603D-1-4		.	-75.2	.	.	-175	.	.	
95-603D-1-7		.	-80.1	.	.	-186	.	.	
95-613-5W-1		.	-84.5	.	.	-177	.	.	
95-613-6X-3		.	-81.6	.	.	-205	.	.	
95-613-6X-5		.	.	.	.	.	2.0	.	
95-613-9X-5		.	.	.	.	.	3.5	.	
95-613-12X-3		.	-81.7	.	.	-184	2.2	.	
95-613-16X-5		.	-84.8	.	.	.	.	.	
95-613--17X-1		.	-83.3	.	.	-192	.	.	
95-613-17X-2		.	-83.5	.	.	-182	1.2	.	
GULF OF CALIFORNIA		.	-76.0	-17.0	1.064	.	.	.	GOLDHABER, 1974 ALEKSEYEV and LEBEDEV, 1975
BLACK SEA 113b		.	-63.7	-16.5	1.050	.	.	.	
BLACK SEA 114a		.	-60.0	-12.3	1.051	.	.	.	
BLACK SEA 109		.	-62.4	9.0	1.076	.	.	.	
BALTIC SEA		10.0	-64.0	-23.0	1.044	.	.	.	LEIN et al., 1980 WHITICAR and SUESS, 1986 (in preparation)
1138-4-1 Antarctic Peninsula		.	-86.5	.	.	-194	-4.5	1.235	
1138-4-2		.	-95.8	.	.	-200	-4.0	1.245	
1138-4-3		.	-89.2	.	.	-198	-3.2	1.243	
1138-4-4		.	-93.2	.	.	-205	-4.1	1.253	
1138-5-1		.	-83.9	.	.	-191	-4.0	1.231	
1145-4-3		-1.3	-97.3	-15.8	1.090	-196	-6.3	1.236	
1145-4-5		-1.3	-76.6	-23.0	1.058	.	-6.4	.	
1145-4-4		-1.3	-101.	.	.	-168	-6.4	1.194	
1145-4-2		-1.3	-91.2	.	.	-187	-6.4	1.222	
1164-1-1		-1.3	-88.1	.	.	-200	-5.0	1.244	
1165-1-1		-1.3	-88.3	.	.	-199	-5.1	1.242	
1166-1-8		-1.3	-63.3	-4.5	1.063	.	-5.0	.	
1166-1-7		-1.3	-87.6	.	.	-211	-5.0	1.261	
1166-1-6		-1.3	-69.7	-18.1	1.079	.	.	.	
1166-1-5		-1.3	-88.6	.	.	-116	-4.8	.	
1166-1-4		-1.3	-88.7	-18.7	1.077	.	.	.	
1166-1-3		-1.3	-97.8	.	.	.	.	.	
1166-1-2		-1.3	-90.7	-6.8	1.092	-200	-3.8	1.246	
1166-1-1		-1.3	-85.4	.	.	-194	-4.0	1.236	
1166-1-0		-1.3	-84.3	.3	1.092	-198	-4.5	1.241	
1186-1-1		-1.3	-109.	-19.4	1.101	-199	-5.2	1.242	
SPECIAL SEDIMENTARY ENVIRONMENTS									
ORCA BASIN		5.7	-104.	-29.7	1.083	.	.	.	SACKETT et al., 1979
ORCA BASIN		5.7	-100.	-29.5	1.078	.	.	.	
ORCA BASIN		5.7	-98.0	-27.7	1.078	.	.	.	
ORCA BASIN		5.7	-96.0	-27.8	1.075	.	.	.	
ORCA BASIN		5.7	-94.0	-24.8	1.076	.	.	.	
ORCA BASIN		5.7	-98.0	-24.6	1.081	.	.	.	
ORCA BASIN		5.7	-90.0	-23.9	1.073	.	.	.	
ORCA BASIN		5.7	-88.0	-24.7	1.069	.	.	.	
ORCA BASIN		5.7	-85.0	-31.3	1.059	.	.	.	
KIVU SEE		26.0	-42.2	.	.	-315	33.0	1.508	SCHOELL, 1983
LAKE KIVU		26.0	-46.5	.	.	-303	86.0	1.558	
LAKE KIVU		26.0	-45.7	.	.	-291	181.	1.666	
LAKE KIVU		26.0	-45.4	.	.	-276	271.	1.756	
LAKE KIVU		26.0	-44.5	.	.	-324	-45.0	1.413	
LAKE KIVU		26.0	-45.0	-2.0	1.045	.	.	.	DEUSER et al., 1973

Table 1d: Biogenic methane data base

SAMPLE	LOCATION	TEMP	$\delta^{13}\text{C}_1$	$\delta^{13}\text{CO}_2$	α_{C}	$\delta\text{D}_{\text{C}_1}$	$\delta\text{D}_{\text{H}_2\text{O}}$	α_{D}	REFERENCE
NAME		$^{\circ}\text{C}$	‰	‰		‰	‰		
BIOGENIC GAS FIELDS									
NIIGATA		44.0	-66.5	-15.8	1.054	-194	-38.2	1.193	NAKAI et al., 1974
MOBARA		26.5	-65.7	-9.6	1.060	-159	-13.1	1.173	
SUWA		17.5	-70.5	-6.0	1.070	-226	-79.4	1.189	
YAMAGATA		17.0	-70.1	-5.0	1.070	.	.	.	NAKAI, 1960
SUWA GAS FIELD	S-1	21.0	-68.5	-9.7	1.063	.	.	.	
SUWA GAS FIELD	S-2	17.5	-74.1	-8.4	1.071	.	.	.	
YAMAGATA GAS F.	Y-1	19.6	-74.7	-8.5	1.072	.	.	.	
YAMAGATA GAS F.	Y-2	17.2	-69.5	-11.9	1.062	.	.	.	
YAMAGATA GAS F.	Y-3	16.2	-74.7	-3.6	1.077	.	.	.	
YAMAGATA GAS F.	Y-4	13.7	-65.5	2.0	1.072	.	.	.	
NIIGATA GAS F.	N-1	20.8	-68.2	-5.8	1.067	.	.	.	
NIIGATA GAS F.	N-2	25.4	-68.4	-5.8	1.067	.	.	.	
NIIGATA GAS F.	N-3	30.7	-67.6	-4.5	1.068	.	.	.	
NIIGATA GAS F.	N-4	45.3	-69.5	-12.2	1.062	.	.	.	SCHOELL, 1980
TOKYO GAS FIELD M		25.8	-74.8	-10.1	1.070	.	.	.	
TOKYO GAS FIELD KR-1		27.0	-69.5	-6.5	1.068	.	.	.	
TOKYO GAS FIELD O		30.6	-67.7	-4.7	1.068	.	.	.	
ITALY/COLLECCHIO 7		26.0	-76.3	.	.	-187	-24.0	1.200	
ITALY/TRESIGALLO 12		50.0	-71.4	.	.	-189	.	.	
JAPAN/OANA YAMAGATA		.	-71.6	-6.9	1.070	.	.	.	
EARL BANE		.	-75.6	-11.3	1.070	.	.	.	
KEITH CLAPPER		.	-71.7	-10.2	1.066	.	.	.	
ERNEST CUNDIFF		.	-75.0	-7.9	1.073	.	.	.	
NAFFZIGER		.	-84.4	.	.	.	.	.	SCHOELL, 1983
MILO HISER		.	-74.8	-5.9	1.074	.	.	.	
INZENHAM 1	Natural Gas,	.	-70.0	.	.	-198	.	.	
INZENHAM 2	S. Germany	.	-67.6	.	.	-192	.	.	
INZENHAM 3		.	-68.1	.	.	-192	-2.0	1.235	
INZENHAM 4		.	-68.9	.	.	-194	-18.0	1.218	
INZENHAM 5		.	-69.5	.	.	-205	-39.0	1.209	
INZENHAM 6		.	-72.2	.	.	-207	.	.	
INZENHAM 7		.	-71.8	.	.	-213	-56.0	1.199	
INZENHAM 8		.	-71.1	.	.	-202	-62.0	1.175	
GAS FIELD 1	NW. Germany	.	-59.6	.	.	-300	-8.8	1.417	(BGR unpublished data)
GAS FIELD 2		.	-61.2	.	.	-304	2.1	1.441	
GAS FIELD 3		.	-61.0	.	.	-304	2.9	1.442	
LABORATORY CULTURE									
SEWAGE SLUDGE		60.0	.	.	.	-348	-54.0	1.451	SCHOELL, 1980
SEWAGE SLUDGE		60.0	.	.	.	-313	9.0	1.469	
SEWAGE SLUDGE		60.0	.	.	.	-260	179.0	1.593	
SEWAGE SLUDGE		60.0	.	.	.	-200	266.0	1.583	
SEWAGE SLUDGE (L.A.)		35	-47.1	4.1	1.054	.	.	.	NISSENBAUM, 1972
SEWAGE SLUDGE		60.0	-49.1	-5.5	1.046	.	.	.	
METHANOL FERMENTATION		30.0	.	.	1.083	.	.	.	ROSENFELD and SILVERMAN, 1959
METHANOL FERMENTATION		23.0	.	.	1.094	.	.	.	
LAND FILL A		20.0	-52.1	16.6	1.072	.	.	.	GAMES and HAYES, 1976
LAND FILL B		20.0	-48.5	-16.1	1.034	.	.	.	
METHANOSARCINA (barkeri)		40.0	-82.6	-42.3	1.045	.	.	.	OVSYANNIKOV and LEBEDEV, 1967
METHANOBACTERIUM (M.o.H)		40.0	-95.6	-42.0	1.061	.	.	.	
M. THERMOAUTOTROPHICUM		65.0	-64.9	-41.4	1.025	.	.	.	
M. THERMOAUTOTR. (5.5 hr)		65.0	-64.5	-39.8	1.025	.	.	.	
" (13 hr)		20.0	-65.1	-37.0	1.045	.	.	.	
" (24 hr)		20.0	-65.0	-27.9	1.061	.	.	.	
" (48 hr)		25.0	-55.3	3.9	1.026	.	.	.	
" (48 hr)		25.0	-50.5	9.8	1.030	.	.	.	
" (168 hr)		60.0	-43.0	.	1.040	.	.	.	
KAMA DAM (max)		.	-58.0	-1.2	1.060	.	.	.	
KAMA DAM (min)		.	-66.0	-1.8	1.069	.	.	.	
CULTURE STUDY		37.0	-44.1	-9.2	1.037	.	.	.	
CULTURE STUDY		37.0	-43.4	-10.0	1.035	.	.	.	
CULTURE STUDY		46.0	-60.1	-28.7	1.033	.	.	.	HAYES & RISATTI (pers. comm.) HEYER et al., 1976
CULTURE STUDY		46.0	-61.9	-28.6	1.035	.	.	.	
METHANOCOCCUS jannaschii		83.0	.	.	1.032	.	.	.	
RUMEN GAS 1		35.0	-58.5	-15.0	1.032	.	.	.	
RUMEN GAS 2		35.0	-60.8	-15.1	1.034	.	.	.	FUCHS et al., 1979
RUMEN GAS 3		35.0	-59.4	-19.4	1.033	.	.	.	
METHANOL FERMENTER 1		30.0	-103.	-34.2	1.077	.	.	.	
METHANOL FERMENTER 2		30.0	-111.	-40.8	1.079	.	.	.	
M. THERMOAUTOTROPHICUM		65.0	-37.6	-3.5	1.035	-206	-71.5	1.170	BELYAEV et al., 1983
M. THERMOAUTOTROPHICUM		65.0	-40.0	-.6	1.041	.	.	.	
CULTURE STUDY		37.0	.	.	1.033	.	.	.	

results are reported relative to PDB and SMOW for $\delta^{13}\text{C}$ and δD respectively in the usual notation:

$$\delta = \left[\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1 \right] \times 10^3 \quad (3)$$

where R is $^{13}\text{C}/^{12}\text{C}$ and D/H respectively.

Carbon isotopes

Bacterial isotope fractionation was recognized as early as 1959 when ROSENFELD and SILVERMAN showed a large carbon isotopic fractionation of 1.094 (23°C) between methanol and methane in a mixed

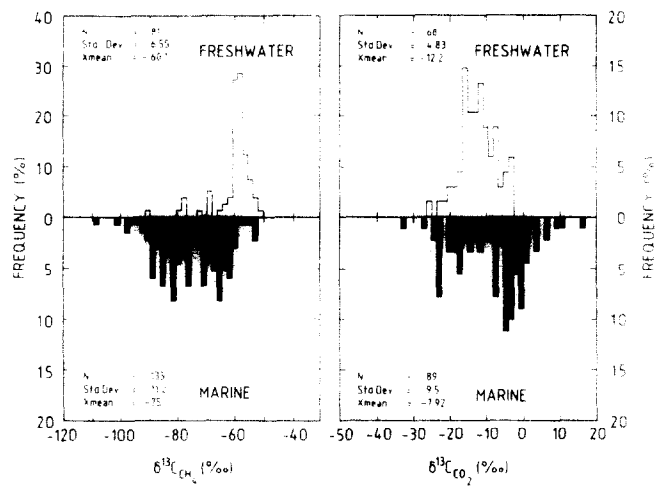


FIG. 2. Histograms of coexisting $\delta^{13}\text{C}$ and CO_2 and biogenic CH_4 from freshwater (open) and marine (shaded) sedimentary environments.

culture study, with methane depleted in $\delta^{13}\text{C}$ relative to PDB up to 88‰. These large depletions of $\delta^{13}\text{C}$ in methane resulting from methanogenesis are shown as a histogram in Fig. 2. The methane $\delta^{13}\text{C}$ is different for the marine and freshwater environments: the methane from the former has a mean of -68‰ and the latter -59‰. A rough boundary could be drawn at -60‰.

The $\delta^{13}\text{C}$ of carbon dioxide in the interstitial water from the methanogenic zone could be separated in Fig. 2 only much more generally into environment types. Freshwater CO_2 was, on average, more negative with a $\delta^{13}\text{C}$ mean of -12‰ which reflects the isotope effect of the terrestrial carbon cycle. The marine CO_2 was more broadly distributed about a mean of -6‰.

The $^{13}\text{C}/^{12}\text{C}$ ratios of dissolved CO_2 in interstitial waters are controlled by a combination of equilibrium and kinetic isotope effects related to atmospheric exchange, precipitation-dissolution, organic matter decomposition, and other diagenetic reactions. This in turn can affect the carbon isotope ratio of biogenic methane. For example, NISSENBAUM *et al.* (1972) and CLAYPOOL and KAPLAN (1974) used the kinetic carbon isotope fractionation by methanogens to generate a wide range of values for both CO_2 and CH_4 depending on the initial $\delta^{13}\text{C}_{\text{CO}_2}$ and the extent of the CO_2 reduction to CH_4 . In spite of this possible range of isotopic CO_2 and CH_4 values, the magnitude of their isotope separation is distinctive for the formation pathway.

The carbon isotopic data of coexisting methane and carbon dioxide pairs, given in Fig. 2, are cross-plotted in Fig. 4. The data segregate into marine and freshwater sedimentary environments. Lines of equal carbon isotopic fractionation (α_c) between methane and carbon dioxide have been drawn as reference from values of $\alpha_c = 1.04$ through 1.09 with the following equation:

$$\alpha_c = \frac{(\delta^{13}\text{C}_{\text{CO}_2} + 10^3)}{(\delta^{13}\text{C}_{\text{CH}_4} + 10^3)} \quad (4)$$

These lines do not necessarily represent equilibrium, rather, they indicate the magnitude of the isotopic separation between these coexisting species. The marine data varied within α_c of 1.05 and 1.09 as expected from the rather wide range of marine carbon isotopic ratios measured. In contrast the freshwater data clustered between α_c of 1.04 and 1.06. The division of marine and non-marine data could be most clearly seen around $\alpha_c = 1.055$ which corresponds to the rough boundary values of -60‰ and -8‰ observed for methane and carbon dioxide in Fig. 2.

The carbon isotope exchange equilibria (α_{cT}) between methane and carbon dioxide has been calculated for various temperatures by CRAIG (1953), BOTTINGA (1969), and RICHET *et al.* (1977) among others. The equations of RICHET *et al.* (1977) have been linearly approximated and are plotted in Fig. 5 using the following equation:

$$10^2 \ln \alpha_{cT} = 2.92(10^3/t) - 2.96 \quad (5)$$

where t is the formation temperature in °K.

The α_c of the CO_2 - CH_4 pairs from the data base are

ENVIRONMENT CODE	special
freshwater	▼ glacial drift
□ swamp	◆ Orca Basin
○ lake	◇ Lake Kivu
marine	accumulations
■ marsh	+ gas reservoirs
▲ shallow water	experimental
● deep water	* culture studies
	△ sewage sludge

FIG. 3. Legend for environment symbols used in following figures.

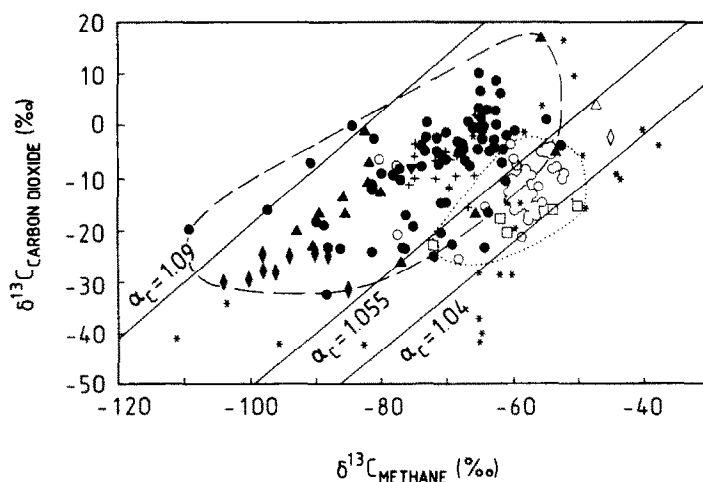


FIG. 4. Cross plot of $\delta^{13}\text{C}$ data from methane and carbon dioxide divided into various sedimentary environments (see Legend). The dashed line is the marine sediment region; the dotted outlines the freshwater sediments. Lines of equal fractionation (Eqn. 4) are drawn for $\alpha_C = 1.04$, 1.055, and 1.09.

also plotted in Fig. 5 as a function of their respective measured sediment temperatures.

With few exceptions, the carbon isotope data of the $\text{CO}_2\text{-CH}_4$ pairs express fractionations lower than if thermodynamic equilibrium had been obtained at the sediment temperatures indicated. The $\text{CO}_2\text{-CH}_4$ fractionation of the marine sediments closely approach the thermodynamic equilibrium line as was noted by GAMES and HAYES (1976). The carbon fractionations in the freshwater environments are further removed from thermodynamic equilibrium and their contrast with the marine sediments does not merely reflect a temperature effect. The carbon isotope exchange rate at shallow sediment temperatures is too slow to permit isotope re-equilibration of the methane that has formed with the carbon dioxide reservoir (CHUNG and SACKETT, 1978; GIGGENBACH, 1982), so that carbon isotope fractionations during methanogenesis are generally described by kinetic isotope effects rather than by isotope exchange equilibria. The magnitude of the carbon isotope effect associated with methane formation in cultures as opposed to those we found in natural en-

vironments have been reported in only a few instances. HEYER *et al.* (1976), GAMES and HAYES (1976), GAMES *et al.* (1978), FUCHS *et al.* (1979), and HAYES and RISATTI (pers. commun.) measured carbon isotope fractionations around 1.032 to 1.035 with extreme values of 1.025 to 1.061 (Table 1). For comparison, ROSENFELD and SILVERMAN (1959), and HEYER *et al.* (1976) reported carbon fractionations between methanol and methane of 1.077 to 1.083 at 30°C and 1.094 at 23°C (Table 1).

$\text{CO}_2\text{-CH}_4$ fractionations estimated by culture studies shown in Fig. 5 are generally lower than those found in either freshwater or marine sediments. In some cases, this may reflect the higher experimental temperatures, or that the difference in rates and types of methanogenesis in laboratories and in natural environments affect isotope discrimination. This points out the difficulties in simulating natural conditions.

As mentioned, one must remain aware of secondary processes which can influence the determination of carbon isotope fractionation factors, such as admixture of methane from other pathways, or from thermogenic hydrocarbons, depletion effects due to limited substrates (closed system kinetics) or microbial oxidation, which can also shift the methane to more positive and the carbon dioxide to more negative values resulting in an overall decrease in the $\text{CO}_2\text{-CH}_4$ fraction.

The Gulf of California DSDP sites 479 and 481 (SCHOELL, 1982) provide an example of admixture of thermogenic hydrocarbons. Methane of biogenic origin only was encountered at site 479. Here the decrease in the carbon isotope fractionation for $\text{CO}_2\text{-CH}_4$ during methanogenesis was a function of the increasing sediment temperature (Table 1, Fig. 5). In contrast, thermogenic hydrocarbons, generated in the proximity of an intrusive dolerite sill at site 481 mix with biogenic methane and significantly lower the apparent $\text{CO}_2\text{-CH}_4$ fractionation (Fig. 5).

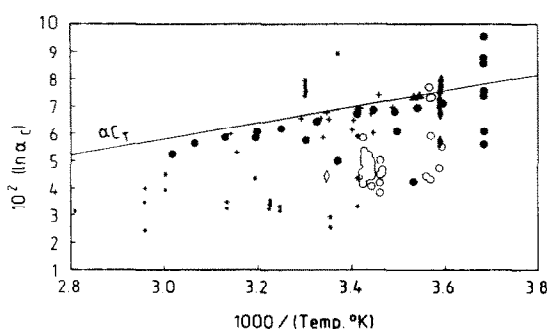


FIG. 5. Temperature dependence of the $\text{CO}_2\text{-CH}_4$ carbon isotope fractionation factor (α_C). The thermodynamic equilibrium fractionation line (α_{CT} Eqn. 5) is also indicated.

Extremely large carbon isotope fractionations are observed (α_C up to 1.10) between CO_2 - CH_4 in the Antarctic sediments (WHITICAR and SUESS, 1986, in preparation). The cold sediment temperature (-1.3°C) is possibly the cause of the higher carbon-12 selectivity in the methane (Table 1) and thus the greater isotope fractionation.

The methane in the anoxic hypersaline Orca Basin sediments are also strongly depleted in carbon-13 (up to -105‰ , SACKETT *et al.*, 1979) and the CO_2 - CH_4 carbon isotope fractionation ($\alpha_C = 1.08$) is consistent with methanogenesis by CO_2 reduction. Unusual, however, is the occurrence of these high methane concentrations in the presence of dissolved sulphate. There is no significant methane consumption in the sulphate zone. Either there is enhanced methanogenesis within the sulphate zone or the methane has vertically migrated. It is not known if the high salinity may have an inhibitory effect on methanogenesis. Experiments by PATEL and ROTH (1977) with NaCl concentrations up to 15‰, showed a species-dependent tolerance or requirement for NaCl. One of the strains investigated, *Methanobacterium* MOH appeared unaffected by the various salt treatment levels. BELYAEV *et al.* (1983) also report inhibition of methanogenesis at salinities over 3‰ for culture studies, but suggested the existence of methanogens which could operate at the elevated salinities they found in the Bondyuzhskoe oil field.

Lake Kivu, a meromictic water body, is a special environment where the $\delta^{13}\text{C}$ methane is very positive (ca. -45‰ , Table 1). Normally this carbon isotope ratio would signify a thermogenic gas source. Studies by TIETZE *et al.* (1980) have shown the methane to be of biogenic origin. The CO_2 is from an abiogenic CO_2 source (-2‰ , DEUSER *et al.*, 1973) and the resulting CO_2 - CH_4 isotope fractionation of 1.045 is in the range with acetate fermentation in freshwater sediments. This interpretation of methanogenesis by acetate fermentation is also supported by the hydrogen isotope data.

The incorporation of the heavy CO_2 signal into the methane could be via CO_2 reduction with an unexplained lower fractionation factor, or alternatively, via a metabolic interaction suggested by ZEIKUS *et al.* (1985) whereby the heavy CO_2 is consumed by acetogens which in turn provide the ^{13}C -enriched acetate for the methanogens.

Hydrogen isotopes

The distinction between freshwater and marine-sourced bacterial methane observed with the carbon isotopes is even more pronounced with the hydrogen isotopes. Marine bacterial methane is narrowly distributed about a mean value for deuterium of -192‰ relative to SMOW (std.dev. $\pm 18.4\text{‰}$) as given in Fig. 6. In contrast, bacterial methane formed in freshwater sediments is strongly depleted in deuterium with an average δD value of -318‰ (std.dev. $\pm 26.5\text{‰}$). Deuterium depletions of up to -400‰ have been reported (ALEKSEYEV, 1980). The means of the hydrogen isotopes from the two populations are separated by 129‰ and, as seen in Fig. 6, there is little overlap. This large isotope separation permits a clear differentiation of the two methanogenic environments. Analogous to the carbon isotopes, the relationship of the hydrogen isotopes between methane and coexisting formation water are pathway dependent.

PINE and BARKER (1956) showed that during reduction of CO_2 to CH_4 in a mixed culture study, the hydrogen of the deuterated water (D_2O) medium was utilized to produce CD_4 . Furthermore, DANIELS *et al.* (1980) conducted similar studies with *M. thermoautotrophicum*, a methanogen which utilizes CO_2 as a sole carbon source, using a strongly deuterated water (D_2O) medium and an initial atmosphere of protium (H_2). They indicated that essentially all the hydrogen incorporated into the methane by the CO_2 reduction pathway was deuterium and originated from the formation water.

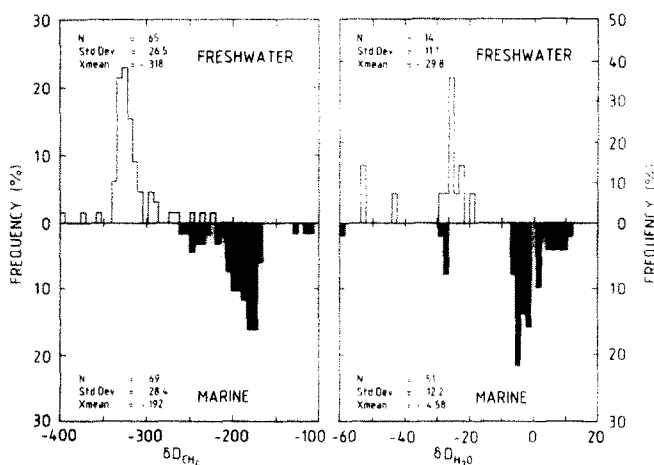


FIG. 6. Histogram of D/H values from methane and formation water, divided into freshwater and marine sedimentary environments.

The hydrogen which is incorporated into the methane is different from that which acts as electron donor, for example by interspecies transfer, in the reduction of the carbon dioxide. FUCHS *et al.* (1979) stated that the hydrogen incorporated into the cells of *M. thermoautotrophicum* had a constant isotope fractionation with respect to the medium water ($\delta D = 1.16$ at 65°C) independent of the gassing rate with hydrogen. Unfortunately in this study neither the hydrogen isotopes of the hydrogen gas nor of the methane formed were measured.

NAKAI *et al.* (1974) and SCHOELL (1980) reported for D/H isotopes a linear function between natural biogenic methane accumulations and their respective formation waters:

$$\delta D_{\text{CH}_4} = \delta D_{\text{H}_2\text{O}} - 160 (\pm 10\%). \quad (6)$$

Based on this direct linear relationship between the methane and the formation water, the latter is interpreted to be the exclusive source of hydrogen in the methane. This is as would be expected for methanogenesis by the CO_2 reduction pathway as presented schematically in Fig. 7. The hydrogen fractionation involved with the attachment of the 4 hydrogen atoms to the carbon during this form of methanogenesis is around 160‰.

In contrast to methanogenesis by CO_2 reduction, PINE and BARKER (1956), and PINE and VISHNIAC (1957) reported that the acetate fermentation pathway involves an intact transfer of the methyl group to the methane, and thus only one externally derived hydrogen atom is required (DANIELS *et al.*, 1980). This distinction in the source of hydrogen is demonstrated by substituting the mean of δD_{CH_4} of -318‰ for methane formed in freshwater environments (Fig. 6) into Eqn.

Sources of hydrogen in bacterial methane

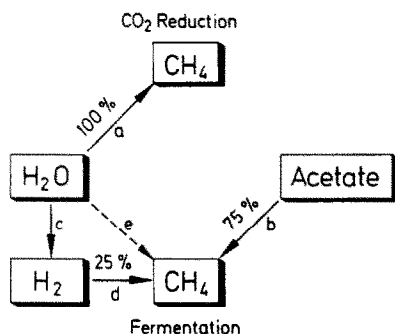


FIG. 7. Schematic diagram of hydrogen transfer for CO_2 reduction and acetate fermentation methanogenic pathways. Formation water supplies the hydrogen for methanogenesis by CO_2 reduction (a), whereas by acetate fermentation, $\frac{3}{4}$ comes directly from acetate (b) and final hydrogen either directly from the formation water (c) or more likely over a hydrogen fractionation step (c + d).

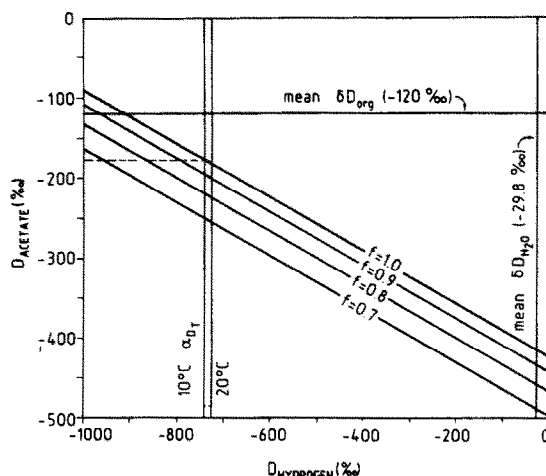


FIG. 8. Relationship of $\delta D_{\text{acetate}}$ to the $\delta D_{\text{hydrogen}}$ (the remaining hydrogen added to methane) using Eqn. 9 for the mean δD_{CH_4} (-318‰) and $\delta D_{\text{H}_2\text{O}}$ (-29.8‰ , shown). Addition of varying proportions of methane from CO_2 reduction ($f = 0.9, 0.8, 0.7$) to acetate fermentation only ($f = 1.0$) are also drawn. For reference, the δD_{org} for freshwater sediments is shown along with α_{DT} , the equilibrium hydrogen isotope fractionation of H_2 from the formation water at 10°C and 20°C .

6 for CO_2 reduction. This yields a calculated value of δD formation water (-158‰) which is much more depleted than was actually determined (-29.8‰).

The hydrogen isotope ratio of methane formed by acetate fermentation can be described relative to the formation water by:

$$\delta D_{\text{CH}_4} = 0.25\delta D_{\text{H}_2\text{O}} + b \quad (7)$$

where $\frac{3}{4}$ of the hydrogen is derived from the methyl group and the formation water is taken to be the ultimate source of the remaining 25% of the hydrogen (Fig. 7).

The value of b is dependent on:

- the hydrogen isotope ratio of the hydrogens in the transferred methyl group of the acetate (or other precursor),
- the hydrogen fractionation associated with this transfer,
- fractionation associated with the addition of the final hydrogen.

The magnitude of (b) in Eqn. 7 calculated using, for example, the freshwater mean δD_{CH_4} and $\delta D_{\text{H}_2\text{O}}$ is -311‰ . The difficulty in determining the correct value for (b), which is variable, is further compounded by contributions of methane through CO_2 reduction which add uncertainty in establishing the true acetate fermentation end member.

Although the hydrogen isotope ratio of the transferred methyl group is unknown, the following can define the range expected. Figure 8 illustrates the dependence of the $\delta D_{\text{acetate}}$ (methyl group) on the δD of the final hydrogen incorporated as is calculated using the mass balance:

$$4(\delta D_{CH_4}) = 3(\delta D_{acetate}) + (\delta D_{hydrogen}) \quad (8)$$

where $\delta D_{hydrogen}$ is the fourth hydrogen incorporated into the methane. For example, inserting the mean δD_{CH_4} of -318‰ for the freshwater methane and assuming that the methane was formed by acetate fermentation only, the line $f = 1.0$ in Fig. 8 is generated. Hydrogen coming directly from the water (mean $\pm 29.8\text{‰}$) without fractionation as indicated by pathway (c) in Fig. 7 would necessitate a $\delta D_{acetate}$ of -414‰ , which is a considerable fractionation with respect to the mean D/H ratio of freshwater organic matter ($\delta D_{org} = -120\text{‰}$, Fig. 8) cited by ESTEP and HOERING (1980) and STILLER and NISSENBAUM (1980). If the hydrogen utilized is, however, first dissociated from the formation water indicated by pathway (c) in Fig. 7, then the former would have a value of -740‰ at 10°C (RICHEL *et al.*, 1977) with respect to the mean water value (Fig. 8). The associated $\delta D_{acetate}$ would be -177‰ in this case (dashed line Fig. 8), or a 57‰ separation relative to the mean δD_{org} . The temperature dependence of the H_2O-H_2 fractionation would not significantly influence the $\delta D_{acetate}$ as shown in Fig. 8.

Methanogenesis in freshwater sediments is stated by numerous investigators (JERIS and MCCARTY, 1965; BELYAEV *et al.*, 1975; CAPPENBERG and JONGEJAN, 1978; MOUNTFORT and ASHER, 1978; and others) to represent a 70:30 mixture of acetate fermentation and CO_2 reduction pathways rather than only the former. Should this be the case, then Eqn. 8 can be modified to include the contribution of CO_2 reduction to acetate fermentation on the δD_{CH_4} measured:

$$\delta D_{CH_4} = f[.75(\delta D_{acetate}) + .25(\delta D_{hydrogen})] + (1-f)(\delta D_{H_2O} - 160) \quad (9)$$

where f is the proportion of methanogenesis varying from acetate fermentation ($f = 1$) to CO_2 reduction ($f = 0$). As the contribution of CO_2 reduction to the methane formed increases, then the calculated $\delta D_{acetate}$ value becomes more depleted in deuterium, assuming the $\delta D_{hydrogen}$ is held constant. Thus at a 70% fermentation ($f = 0.7$, Fig. 8) and a $\delta D_{hydrogen}$ via pathway (c) (Fig. 7) of -740‰ a value of $\delta D_{acetate}$ would be -250‰ .

Hydrogen isotope data pairs of bacterial methane and the coexisting formation water are less commonly reported in the literature. Those available (Table 1) are plotted in Fig. 9. Methane from marine sediments cluster close to the region of methanogenesis by CO_2 reduction as defined by the line (Eqn. 6) drawn in Fig. 9. This methane generally has a H_2O-CH_4 fractionation (δ_D) between 1.2 and 1.3. Variations from the line may be due to a minor addition of methane through acetate fermentation (Fig. 1). Alternatively, the relationship of δD_{CH_4} to δD_{H_2O} could be revised as the function:

$$\delta D_{CH_4} = \delta D_{H_2O} - 180 \pm 10\text{‰}. \quad (10)$$

Reservoired bacterial methane from marine sediment sources in Japan, Italy and S. Germany also plot in the region of methanogenesis by CO_2 reduction.

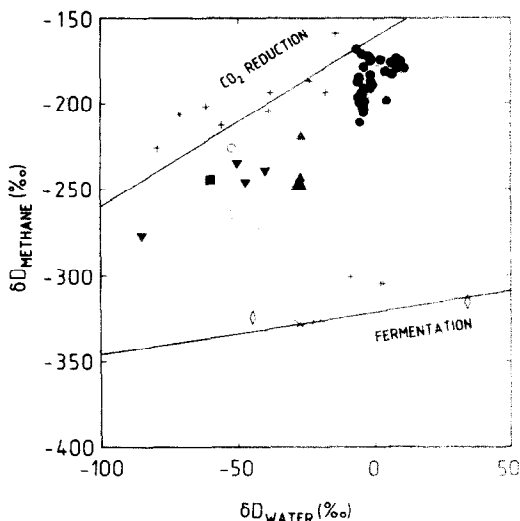


FIG. 9. Cross plot of methane and formation water δD coexisting pairs delineating the regions of freshwater and marine sediments. The lines describing methanogenesis by CO_2 reduction (Eqn. 6) and acetate fermentation (Eqn. 7) are shown.

In contrast, the bacterial methane from freshwater sediments in Fig. 9 plot removed from the region describing CO_2 reduction. The coexisting hydrogen pairs of Wurmsee, Altwarmbüchen Moor, and Lake Kivu have H_2O-CH_4 fractionations (α_D) greater than 1.40. The line in Fig. 9 describing acetate fermentation fitting this data and that found experimentally with sewage sludges (SCHOELL, 1980) is:

$$\delta D_{CH_4} = 0.25\delta D_{H_2O} - 321\text{‰}. \quad (11)$$

As discussed above, the range in values and thus position of the line can be related to differences in $\delta D_{acetate}$, $\delta D_{hydrogen}$ (the final hydrogen incorporated), or in a mixed situation, the proportion of the respective methanogenic pathways. Interestingly, some of the freshwater environments (Brake, Glacial Drift, Volo Bog in Table 1) indicate major contributions by CO_2 reduction. Thus this methanogenic pathway may also be significant in freshwater environments (Fig. 1) though probably in the later stages due to the lower rates of methanogenesis by CO_2 reduction.

The isotope separations for both carbon (CO_2-CH_4) and hydrogen (H_2O-CH_4) have been shown to be individually distinctive of the methanogenic pathway and thus indirectly of the environment of formation. Although the current number of samples for which both fractionation factors (α_C , α_D) are available is limited, by combining them in Fig. 10 the two types of methane formation CO_2 reduction and acetate fermentation are readily distinguished. The former has a larger carbon fractionation ($\alpha_C = 1.07$) and a lower hydrogen fractionation (α_D ca. 1.2), whereas the acetate fermentation is the opposite (α_C ca. 1.045 and α_D ca. 1.45). This clear division of biogenic methane types illustrates the importance of measuring not only the carbon isotopes

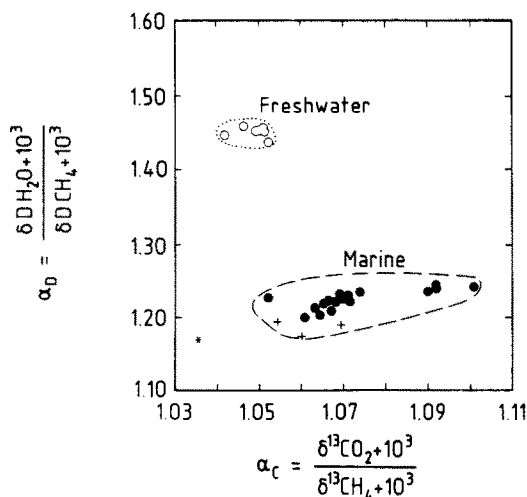


FIG. 10. Combination of carbon and hydrogen isotope fractionations (α_c and α_D) for methane, carbon dioxide and formation water. The regions of CO_2 reduction and acetate fermentation are clearly differentiated using the coexisting $\text{CO}_2\text{-CH}_4$ and $\text{H}_2\text{O-CH}_4$ pairs.

of methane, but also the hydrogen isotopes, as well as the isotopes of the coexisting CO_2 and H_2O .

Genetic characterization of biogenic methane accumulations

The source of a bacterial methane reservoir, *i.e.* whether it was generated in freshwater or marine sediments, can be determined with the aid of carbon and hydrogen isotopes. The measurement of only methane carbon isotopes on reservoir natural gases is often insufficient to delineate the source. This difficulty is demonstrated using the 129 bacterial gas reservoirs listed by RICE and CLAYPOOL (1981). A histogram of their data in Fig. 11 suggests no separation between environment types. Referring to the superimposed curves from the freshwater and marine $\delta^{13}\text{C}_{\text{CH}_4}$ frequency distributions from Fig. 2a, a rough value of -60‰ for unaltered biogenic methane could perhaps be used as a reference boundary. A more reliable interpretation, however, requires information on the hydrogen isotopes and the $\delta^{13}\text{C}$ of the coexisting CO_2 .

The distinction between biogenic and thermogenic natural gas is also of importance in hydrocarbon exploration. Again the singular use of either carbon or hydrogen isotopes can lead to ambiguous interpretations. These uncertainties can frequently be resolved by the inclusion of additional isotopic or sometimes molecular composition data. Figure 12 is a revised $\delta^{13}\text{C}$ vs. δD diagram after SCHOELL (1980) which incorporates the now expanded field for bacterial methane, particularly identifying the freshwater environments.

This diagram distinguishes the various methane sources. For example, a natural gas with a methane $\delta^{13}\text{C}$ value of -52‰ could previously have been identified as a gas from a mixed source. With the knowledge

that the methane δD is -350‰ , it can be properly classified as a freshwater biogenic gas.

Mixing or alteration effects can still contribute difficulties to the genetic characterization of natural gases. Further effort is required to increase our understanding of the isotopic behaviour of natural gases under these influences. It must be emphasized that any interpretation should incorporate not only the geochemical information available, but also the geologic and geophysical situation as well.

CONCLUSIONS

Biogenic methane from freshwater and marine environments can be distinguished with stable carbon and hydrogen isotopes. The primary methanogenic pathways, acetate-type fermentation and CO_2 reduction, can be characterized by combining the isotope data of methane with the data from the coexisting dissolved CO_2 and formation water.

In marine sediments the methane formed by CO_2 reduction is often more depleted in ^{13}C and has a greater carbon isotope separation ($\text{CO}_2\text{-CH}_4$) than for methanogenesis by acetate fermentation in freshwater sediments.

Methane formed by the transfer of a methyl group is more depleted in deuterium than that formed by reduction of CO_2 . Hydrogen for the latter is incorporated directly from the formation water with a constant fractionation. In contrast, during acetate fermentation only $1/4$ of the hydrogen is derived from the formation water, probably *via* an intermediate $\text{H}_2\text{O-H}_2$ dissociation step.

Biogenic methane formed by CO_2 reduction and acetate-type fermentation pathways can be present in both freshwater and marine environments, however, the predominance of the latter in freshwater sediments is due to the availability of methyl from precursors such as acetate. In marine sediments, methanogenesis is limited in sediments with dissolved sulphate, due to competition for hydrogen and acetate-type precursors.

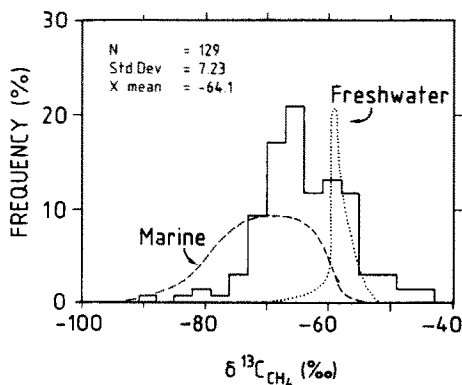


FIG. 11. Histogram of methane $\delta^{13}\text{C}$ in natural gas reservoirs, (solid) after RICE and CLAYPOOL (1981). Freshwater (dotted) and marine (dashed) histograms from Fig. 3 are superimposed as a rough environmental indicator.

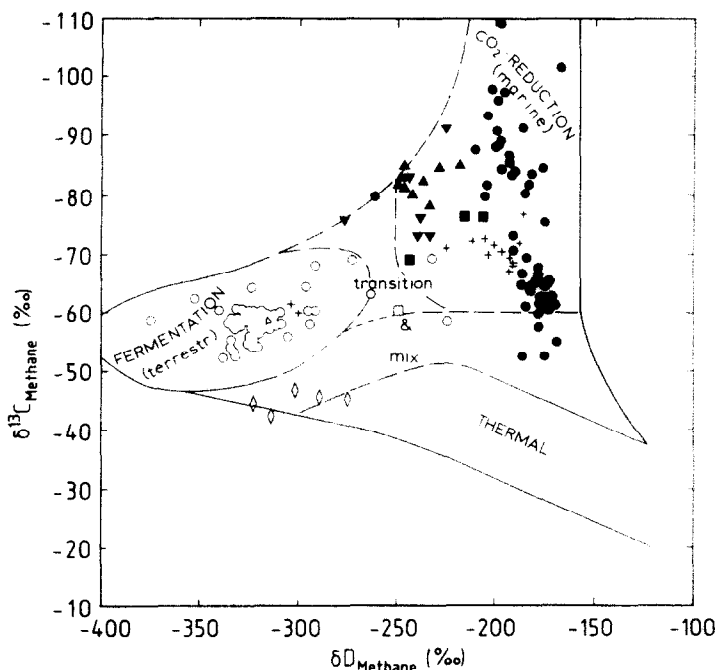


FIG. 12. Natural gas genetic classification diagram using δC and δD of methane. The biogenic methane strongly depleted in deuterium through acetate-type fermentation can be distinguished from methanogenesis by CO_2 reduction and from thermogenic methane.

The generalized systematics inherent to a large data base should be further refined by detailed examination of specific natural settings, complemented by laboratory studies.

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REFERENCES

- ABRAM J. W. and NEDWELL D. B. (1978a) Inhibition of methanogenesis by sulfate-reducing bacteria competing for transferred hydrogen. *Arch. Microbiol.* **117**, 89–92.
- ABRAM J. W. and NEDWELL D. B. (1978b) Hydrogen as a substrate for methanogenesis and sulfate reduction in anaerobic salt marsh sediment. *Arch. Microbiol.* **117**, 93–97.
- ALEKSEYEV F. A. (1980) New isotopic radiogeochemical data on petroleum production. *Int. Geol. Rev.* **22**, No. 8, 899–904.
- ALEKSEYEV F. A. and LEBEDEV W. C. (1975) The carbon isotope composition of carbon dioxide and methane in the bottom sediments of the Black Sea. In *Dispersed Gases and Biochemical Conditions of Sediments and Rocks*, Nedza, Moscow.
- BALCH W. E., FOX G. E., MAGRUM L. J., WOESCE C. R. and WOLFE R. S. (1979) Methanogens: reevaluation of a unique biological group. *Microbiol. Rev.* **43**, 260–296.
- BARBER L. (1974) Methane production in sediments of Lake Wingra. Ph.D. thesis, Univ. Wisconsin, Madison, 545 p.
- BARKER H. A. (1936) On the biochemistry of the methane fermentation. *Arch. Mikrobiol.* **7**, 420–438.
- BARNES R. O. and GOLDBERG E. D. (1976) Methane production and consumption in anoxic marine sediments. *Geology* **4**, 297–300.
- BELYAEV S. S., FINKELSTEIN Z. I. and IVANOV M. V. (1975) Intensity of bacterial methane formation in ooze deposits of certain lakes. *Microbiol.* **44**, 272–275.
- BELYAEV S. S., LAURINAVICHUS K. S. and GAYTAN V. I. (1977) Modern microbiological formation of methane in the quaternary and pliocene rocks of the Caspian Sea. *Geochem. Intern.* **4**, 172–176.
- BELYAEV S. S., WOLKIN R., KENEALY W. R., DENIRO M. J., EPSTEIN S. and ZEIKUS J. G. (1983) Methanogenic bacteria from the Bondyuzhskoe Oil Field: General characterization and analysis of stable-carbon isotopic fractionation. *Appl. Environ. Microbiol.* **45**, 691–697.
- BOSSARD P. (1981) Der Sauerstoff- und Methanhaushalt im Lungernsee. Ph.D. thesis, E.T.H. Zürich, 71 p.
- BOTTINGA Y. (1969) Calculated fractionation factors for carbon and hydrogen isotope exchange in the system calcite-carbon dioxide-graphite-methane-hydrogen-water vapour. *Geochim. Cosmochim. Acta* **33**, 49–64.
- BUSWELL A. M. and SOLO F. W. (1948) The mechanism of the methane fermentation. *J. Amer. Chem. Soc.* **70**, 1179–1180.
- CAPPENBERG TH. E. (1974) Interrelations between sulfate-reducing and methane-producing bacteria in bottom deposits of a freshwater lake, I. Field observations. *J. Microbiol. Serol.* **40**, 285–295.
- CAPPENBERG TH. E. (1975) A study of mixed continuous cultures of sulfate-reducing and methane-producing bacteria. *Microbiol. Ecol.* **2**, 60–72.
- CAPPENBERG TH. E. and PRINS R. A. (1974) Interrelations between sulfate-reducing and methane-producing bacteria in bottom deposits of a fresh-water lake III. Experiments with ^{14}C -labeled substrates. *J. Microbiol. Serol.* **40**, 457–469.

- CAPPENBERG TH. E. and JONGEJAN E. (1978) Microenvironments for sulfate reduction and methane production in freshwater sediments. In *Environmental Biogeochemistry and Geomicrobiology* (ed. W. E. KRUMBEIN), pp. 129–138. Ann Arbor Science, Ann Arbor.
- CHUNG H. M. and SACKETT W. M. (1978) Carbon isotope effects during the pyrolytic formation of methane from coal. *Short Papers of 4th Int. Conf. Geoch. Cosmoch. Isotope Geol.* (ed. R. E. ZARTMANN), *USGS Open File Report* 78, 701.
- CLAYPOOL G. E. and KAPLAN I. R. (1974) The origin and distribution of methane in marine sediments. In *Natural Gases in Marine Sediments* (ed. I. R. KAPLAN), pp. 99–139. Plenum Press, New York.
- CLAYPOOL G. E. and THRELKELD C. N. (1983) Anoxic diagenesis and methane generation in sediments of the Blake Outer Ridge, DSDP Site 533, Leg 76. In *Initial Reports Deep Sea Drilling Project*, Vol. 76 (eds. R. A. SHERIDAN and F. M. GRADSTEIN), pp. 391–402. U.S. Govt. Printing Office, Washington.
- CLAYPOOL G. E., PRESLEY B. J. and KAPLAN I. R. (1973) Gas analysis in sediment samples from legs 10, 11, 13, 14, 15, 18, 19. In *Initial Reports Deep Sea Drilling Project*, Vol. 19 (eds. J. S. CREAGER and D. W. SCHOLL), pp. 879–884. U.S. Govt. Printing Office, Washington.
- COLEMAN D. D. (1976) The origin of drift-gas deposits as determined by radiocarbon dating of methane. *Proc. 9th Intl. Radiocarbon Conf. L.A.*, 20–26.
- CRAIG H. (1953) The geochemistry of the stable carbon isotopes. *Geochim. Cosmochim. Acta* 3, 53–92.
- DANIELS L., FULTON G., SPENCER R. W. and ORME-JOHNSON W. H. (1980) Origin of hydrogen in methane produced by methanobacterium thermoautotrophicum. *J. Bacteriology* 141, 694–698.
- DAVIS J. B. and YARBROUGH H. E. (1966) Anaerobic oxidation of hydrocarbons by desulfovibrio desulfuricans. *Chem. Geol.* 1, 137–144.
- DEUSER W. G., HARVEY G. R., DEGENS E. T. and RUBIN M. (1973) Methane in Lake Kivu: new data on its origin. *Science* 181, 51–54.
- DEVOL A. H. (1983) Methane oxidation rates in the anaerobic sediments of Saanich Inlet. *Limnol. Oceanog.* 28, 738–742.
- DOOSE P. R. (1980) The bacterial production of methane in marine sediments. Ph.D. thesis, U.C.L.A. 240 p.
- ESTEP M. F. and HOERING T. C. (1980) Biochemistry of the stable hydrogen isotopes. *Geochim. Cosmochim. Acta* 44, 1197–1206.
- FABER E. and STAHL W. (1983) Analytic procedure and results of an isotope geochemical surface survey in an area of the British North Sea. In *Petroleum Geochemistry and Exploration of Europe* (ed. J. BROOKS), pp. 51–63. Blackwell Sci. Pub., London.
- FRIEDMANN I. and HARDCASTLE K. (1973) Interstitial water studies, leg. 15—Isotopic composition of water. In *Initial Reports Deep Sea Drilling Project*, Vol. 20 (eds. B. C. HEEZEN and I. D. MACGREGOR), pp. 901–903. U.S. Govt. Printing Office, Washington.
- FROELICH P. N., KLINKHAMMER G. P., BENDER M. K., LUEDTKE N. A. et al. (1979) Early oxidation of organic matter in pelagic sediments of the eastern equatorial Atlantic: suboxic diagenesis. *Geochim. Cosmochim. Acta* 43, 1075–1090.
- FUCHS G. D., THAUER R., ZIEGLER H. and STICHLER W. (1979) Carbon isotope fractionation by Methanobacterium thermoautotrophicum. *Arch. Microbiol.* 120, 135–139.
- GAMES L. M. and HAYES J. M. (1976) On the mechanisms of CO₂ and CH₄ production in natural anaerobic environments. In *Proceedings of the 2nd Intl. Symp. on Environmental Biogeochemistry* (ed. J. O. NRIAGU), pp. 51–73. Science Press, Ann Arbor, Mich.
- GAMES L. M., HAYES J. M. and GUNSALAS R. D. (1978) Methane producing bacteria: natural fractionations of the stable carbon isotopes. *Geochim. Cosmochim. Acta* 42, 1295–1297.
- GIGGENBACH W. F. (1982) Carbon-13 exchange between CO₂ and CH₄ under geothermal conditions. *Geochim. Cosmochim. Acta* 46, 159–165.
- GOLDHABER M. B. (1974) Equilibrium and dynamic aspects of the marine geochemistry of sulfur. Ph.D. thesis, Univ. California, L.A.
- HEYER L., HÜBNER H. and MAAß I. (1976) Isotopenfraktionierung des Kohlenstoffs bei der mikrobiellen Methanbildung. *Isotopenpraxis* 12, 202–205.
- HIPPE H., CASPARI D., FIEBIG K. and GOTTSCHALK G. (1979) Utilization of trimethylamine and other *n*-methyl compounds for growth and methane formation by Methanoscarcina barkeri. *Proc. Natl. Acad. Sci.* 76, 494–498.
- HOPPE-SEYLER F. (1887) Die Methangaeährung der Essigsäure. *Hoppe-Seyler's Z. Physiol. Chem.* 11, 561–568.
- INGVORSEN K. and BROCK T. D. (1982) Electron flow via sulfate reduction and methanogenesis in the anaerobic hypolimnion of Lake Mendota. *Limnol. Oceanog.* 27, 559–564.
- JERIS J. S. and MCCARTY P. L. (1965) The biochemistry of methane fermentation using ¹⁴C tracers. *J. Water. Pollut. Contr. Fed.* 37, 178–192.
- KING G. M. and WIEBE W. J. (1978) Methane release from soils of a Georgia salt marsh. *Geochim. Cosmochim. Acta* 42, 343–348.
- KING G. M., KLUG M. J. and LOVLEY D. R. (1983) Metabolism of acetate, methanol and methylated amines in intertidal sediments of Lowes Cove, Maine. *Appl. Environ. Microbiol.* 45, 1848–1853.
- KOSIUR D. R. and WARFORD A. L. (1979) Methane production and oxidation in Santa Barbara Basin Sediments. *Estuar. Coast. Mar. Sci.* 8, 379–385.
- KOYAMA T. (1964) Gaseous metabolism in lake sediments and paddy soils. In *Advances in Organic Geochemistry* (eds. U. COLUMBO and G. D. HOBSON), pp. 363–375. MacMillan Co., New York.
- LAWRENCE A. W. and MCCARTY P. L. (1965) The role of sulphide in preventing heavy metal toxicity in anaerobic treatment. *Water Poll. Contr. Fed. J.* 37, 392–409.
- LAWRENCE A. W., MCCARTY P. L. and GUERING F. J. A. (1964) In *Proc. 19th Indiana Waste Conf., Eng. Bull.* 117, pp. 343–357. Purdue Univ. Ext. Ser.
- LEBEDEW W. C., OWSJANNIKOW W. M., MOGILEWSKIJ G. A. and BOGDANOW W. M. (1969) Fraktionierung der Kohlenstoffisotope durch mikrobiologische Prozesse in der biochemischen Zone. *Angew. Geol.* 12, 621–624.
- LEIN Y. YU., NAMSARAEV B. B., TROTSYUK V. YA. and IVANOV M. V. (1981) Bacterial methanogenesis in Holocene sediments of the Baltic Sea. *Geomicrobiology J.* 2, 299–315.
- LOVLEY D. R., DWYER D. F. and KLUG M. J. (1982) Kinetic analysis of competition between sulfate reducers and methanogens for hydrogen in sediments. *Appl. Environ. Microbiol.* 43, 1373–1379.
- LYON G. (1973) Interstitial water studies, leg. 15—Chemical and isotopic composition of gases from Cariaco Trench sediments. In *Initial Reports Deep Sea Drilling Project*, Vol. 20 (eds. B. C. HEEZEN and I. D. MACGREGOR), pp. 773–774. U.S. Govt. Printing Office, Washington.
- MAH R. A., WARD D. M., BARESI L. and GLOSS T. L. (1977) Biogenesis of methane. *Ann. Review Microbiology* 31, 309–341.
- MAKONGON Y. F., TSAREV V. I. and CHERSKIY N. V. (1972) Formation of large natural gas fields in zones of permanently low temperatures. *Akad. Narsk SSSR Doklady* 205, 215–218.
- MARTENS C. S. (1976) Control of methane sediment-water transport by macroinfaunal irrigation in Cape Lookout Bight, North Carolina. *Science* 192, 998–1000.
- MARTENS C. S. (1982) Methane production, consumption and transport in the interstitial waters of coastal marine sediments. In *Dynamic Environment of the Ocean Floor* (eds. K. A. FANNING and F. T. MANHEIM), pp. 187–202. Lexington Books, Toronto.

- MARTENS C. S. and BERNER R. A. (1977) Interstitial water chemistry of anoxic Long Island Sound sediments I. Dissolved gases. *Limnol. Oceanogr.* **22**, 10–25.
- MOUNTFORT D. O. and ASHER R. A. (1978) Changes in proportions of acetate and carbon dioxide used as methane precursors during the anaerobic digestion of bovine waste. *Appl. Environ. Microbiol.* **35**, 648–654.
- MOUNTFORT D. O., ASHER R. A., MAYS E. L. and TIEDJE J. U. (1980) Carbon and electron flow in mud and sandflat intertidal sediments at Delaware Inlet, Nelson, N. Z., *Appl. Environ. Microbiol.* **39**, 686–694.
- MYLROIE R. L. and HUNGATE R. E. (1955) Experiments on the methane bacteria in sludge. *Can. J. Microbiol.* **1**, 55–64.
- MÜLLER P. J. and SUSS E. (1979) Productivity, sedimentation rate and sedimentary organic matter in oceans I. Organic carbon preservation. *Deep-Sea Res.* **26**, 1347–1362.
- NAKAI N. and NOBUYUKI N. (1960) Carbon isotope fractionation of natural gas in Japan. *J. Earth Sci. (Nagoya Univ.)* **8**, 174–180.
- NAKAI M., YOSHIDA Y. and ANDO N. (1974) Isotopic studies on oil and natural gas fields in Japan. *Chikyakaya (Geochem.)* **7/8**, No. 1, 87–98.
- NEDWELL D. B. (1982) The cycling of sulphur in marine and freshwater sediments. In *Sediment Microbiology* (eds. D. B. NEDWELL and C. M. BROWN), pp. 73–145. Academic Press, London.
- NIKAIDO N. (1977) On the relation between methane production and sulfate reduction in bottom muds containing sea water sulphate. *Geochem. J.* **11**, 199–206.
- NISSENBAUM A., PRESLEY B. J. and KAPLAN I. R. (1972) Early diagenesis in a reducing fjord, Saanich Inlet, B. C., I. Chemical and isotopic changes in major components of interstitial water. *Geochim. Cosmochim. Acta* **36**, 1007–1027.
- NOTTINGHAM P. M. and HUNGATE R. E. (1969) Methanogenic fermentation of benzoate. *J. Bacteriol.* **98**, 1170–1172.
- OANA S. and DEEVEY E. S. (1960) Carbon-13 in lake waters and its possible bearing on paleolimnology. *Amer. J. Sci.* **258A**, 253–272.
- OPPERMANN R. A., NELSON W. O. and BROWN R. E. (1961) In vivo studies of methanogenesis in the bovine rumen: dissimilation of acetate. *J. Gen. Microb.* **25**, 103–111.
- OREMLAND R. S. (1985) The biochemistry of methanogenic bacteria (submitted).
- OREMLAND R. S. and TAYLOR B. F. (1975) Microbial production of methane in a tropical seagrass community. *Ann. Mtg. Amer. Soc. Microbiol., Abstr.* **N8**, 185.
- OREMLAND R. S. and TAYLOR B. F. (1978) Sulfate reduction and methanogenesis in marine sediment. *Geochim. Cosmochim. Acta* **2**, 209–214.
- OVSYANNIKOV V. M. and LEBEDEV V. S. (1967) Isotopic composition of carbon in gases of biochemical origin. *Geochem. Intl.* **43**, 453–458.
- PANGANIBAN A. T., PATT T. E., HART W. and HANSON R. S. (1979) Oxidation of methane in the absence of oxygen in lake water samples. *Appl. Environ. Microbiol.* **37**, 303–309.
- PATEL G. B. and ROTH L. A. (1977) Effect of sodium chloride on growth and methane production of methanogens. *Can. J. Microbiol.* **23**, 893–897.
- PINE M. J. and BARKER H. A. (1956) Studies on methane fermentation XII. The pathway of hydrogen in the acetate fermentation. *J. Bacteriol.* **71**, 644–648.
- PINE M. J. and VISHNIAC W. (1957) The methane fermentation of acetate and methanol. *J. Bacteriol.* **73**, 736–742.
- POLCIN S., MARSH L. and OREMLAND R. S. (1981) Methanogenesis and sulfate-reduction: A mechanism for simultaneity. *E.O.S., Trans Amer. Geophys. Union* **62**, No. 45, 925.
- REEBURGH W. S. (1976) Methane consumption in Cariaco Trench waters and sediments. *Earth Planet. Sci. Lett.* **28**, 337–344.
- REEBURGH W. S. (1980) Anaerobic methane oxidation: Rate depth distributions in Skan Bay sediments. *Earth Planet. Sci. Lett.* **47**, 345–352.
- REEBURGH W. S. and HEGGIE D. T. (1974) Depth distributions of gases in shallow water sediments. In *Natural Gases in Marine Sediments* (ed. I. R. KAPLAN), pp. 27–45. Plenum Press, New York.
- REEBURGH W. S. and HEGGIE D. T. (1977) Microbial methane consumption reactions and their effects on methane distributions in freshwater and marine environments. *Limnol. Oceanogr.* **22**, 1–9.
- RICE D. D. and CLAYPOOL G. E. (1981) Generation, accumulation and resource potential of biogenic gas. *Bull. AAPG* **65**, 5–25.
- RICHT P., BOTTINGA Y. and JAVOY M. (1977) A review of hydrogen, carbon, nitrogen, oxygen, sulphur and chlorine stable isotope fractionation among gaseous molecules. *Ann. Rev. Earth Planet. Sci.* **5**, 65–110.
- ROSENFELD W. D. and SILVERMANN S. R. (1959) Carbon isotope fractionation in bacterial production of methane. *Science* **130**, 1658–1659.
- ROSSOLIMO L. L. (1935) Die Boden-Gasausscheidung und das Sauerstoffregime der Seen. *Int. Ver. Theor. Angew. Limnol. Verh.* **7**, 539–561.
- RUDD J. W. M., HAMILTON R. D. and CAMPBELL N. E. R. (1974) Measurements of the microbial oxidation of methane in lake water. *Limnol. Oceanogr.* **19**, 519–524.
- SACKETT W. M., BROOKS J. M., BERNARD B. B., SCHWAB C. R. S. and CHUNG H. (1979) A carbon inventory for Orca Basin brines and sediments. *Earth Planet. Sci. Lett.* **44**, 73–81.
- SANDBECK K. A. and WARD D. M. (1981) Fate of immediate methane precursors in low-sulfate, hot-spring algal-bacterial mats. *Appl. Environ. Microbiol.* **41**, 775–782.
- SCHOELL M. (1980) The hydrogen and carbon isotopic composition of methane from natural gases of various origins. *Geochim. Cosmochim. Acta* **44**, 649–661.
- SCHOELL M. (1982) Stable isotopic analysis of interstitial gases in Quaternary sediments from the Gulf of California. In *Initial Reports Deep Sea Drilling Project*, Vol. 64 (eds. J. R. CURRAY and D. G. MOORE), pp. 815–817. U.S. Govt. Printing Office, Washington.
- SCHOELL M. (1983) Wasserstoff- und Kohlenstoff-Isotopenvariationen in organischen Substanzen, Erdölen und Erdgasen. Habilitationsschrift, Univ. Bochum, 248p.
- SCHUBEL J. R. (1974) Gas bubbles and the acoustically impenetrable or turbid character in some estuarine sediments. In *Natural Gases in Marine Sediments* (ed. I. R. KAPLAN), pp. 275–297. Plenum Press, N.Y.
- SCHÖNHEIT P., KRISTIANSSON J. K. and THAUER R. K. (1982) Kinetic mechanism for the ability of sulfate reducers to out-compete methanogens for acetate. *Arch. Microbiol.* **132**, 285–288.
- SMITH P. H. and MAH R. A. (1966) Kinetics of acetate metabolism during sludge digestion. *Appl. Microbiol.* **14**, 368–371.
- SMITH M. R. and MAH R. A. (1978) Growth and methanogenesis by *Methanosarcina* strain 227 on acetate and methanol. *Appl. Environ. Microbiol.* **36**, 870–879.
- SMITH M. R. and MAH R. A. (1980) Acetate as a sole carbon and energy source for growth of *Methanosarcina* strain 227. *Appl. Environ. Microbiol.* **39**, 993–999.
- STADTMANN T. C. and BARKER H. A. (1949) Studies on the methane fermentation, VI. Tracer experiments on the mechanism of methane formation. *Arch. Biochem. Biophys.* **21**, 256–264.
- STILLER M. and NISSENBAUM A. (1980) Variations of stable hydrogen isotopes in plankton from a freshwater lake. *Geochim. Cosmochim. Acta* **44**, 1099–1101.
- STRAYER R. F. and TIEDJE J. M. (1978) *In situ* methane production in a small, hypereutrophic hard water lake: Loss of methane from sediments by vertical diffusion and ebullition. *Limnol. Oceanogr.* **23**, 1201–1206.

- TAKAI Y. (1970) The mechanism of methane fermentation in flooded paddy soil. *Soil Sci. Plant Nutr.* **16**, 238–244.
- TIETZE K., GEYH M., MÜLLER H., SCHRÖDER L., STAHL W. and WEHNER H. (1980) The genesis of the methane in Lake Kivu (Central Africa). *Geol. Rundschau* **69**, 452–472.
- VOYTOV G. I., LEBEDER V. S., BLOKHINA G. G. and CHEREVICHNAYA L. F. (1975) Chemical and isotopic composition of subsoil and marsh gases of the Kola Tundras. *Doklady Akad. Nauk. SSSR* **224**, 186–188.
- WASSERBURG G. J., MAZOR J. E. and ZARTMANN R. E. (1963) Isotopic and chemical composition of some terrestrial natural gases. In *Earth Science and Meteorites* (eds. J. GEISS and E. D. GOLDBERG), pp. 219–240. North Holland Publ., Amsterdam.
- WEIMER P. J. and ZEIKUS J. G. (1978) Acetate metabolism in *Methanosarcina barkeri*. *Arch. Microbiol.* **119**, 175–182.
- WHELAN T. (1974) Methane carbon dioxide and dissolved sulfate from interstitial water of coastal marsh sediments. *Estuar. Coast. Mar. Sci.* **2**, 407–415.
- WHELAN T., BERNARD B. B. and BROOKS J. M. (1978) Carbon isotope variations in total carbon dioxide and methane from interstitial waters of nearshore sediments. In *Stable Isotopes in Earth Science* (ed. B. W. ROBINSON), pp. 39–47. Dept. Sci. Indiana. Res. Bull. 220. Indiana.
- WHITICAR M. J. (1978) Relationships of interstitial gases and fluids during early diagenesis in some marine sediments. Ph.D. dissertation, Univ. Kiel, 153p.
- WHITICAR M. J. (1982) The presence of methane bubbles in the acoustically turbid sediments of Eckernförder Bay, Baltic Sea. In *Dynamic Environment of the Ocean Floor* (eds. K. A. FANNING and F. T. MANHEIM), pp. 219–235. Lexington Books, Mass.
- WHITICAR M. J. and WERNER F. (1981) Pockmarks: Submarine vents of natural gas or freshwater seeps. *Geo-Marine Letters* **1**, 193–199.
- WHITICAR M. J. and FABER E. (1985) Methane oxidation in marine and limnic sediments and water columns. 12th Intl. Mtg. Org. Geochem., Juelich 16–20, 1985 abstract, p. 20. In *Advances in Organic Geochemistry* (eds. D. LEYTHAUSER and J. RULLKÖTTER) (in press).
- WHITICAR M. J. and FABER E. (1986) Carbon and hydrogen isotopes on gas samples from Leg 95 Sites 603 D and 613. In *Initial Reports Deep Sea Drilling Project*, Vol. 95, U.S. Govt. Printing Office, Washington (in press).
- WINFREY M. R. and ZEIKUS J. G. (1977) Effects of sulphate on carbon and electron flow during microbial methanogenesis in freshwater sediments. *Appl. Environ. Microbiol.* **33**, 275–281.
- WINFREY M. R. and WARD D. M. (1983) Substrates for sulfate reduction and methane production in intertidal sediments. *Appl. Environ. Microbiol.* **45**, 193–199.
- WINFREY M. R., NELSON D. R., KLEVICKIS S. C. and ZEIKUS J. G. (1977) Association of hydrogen metabolism with methanogenesis in Lake Mendota sediments. *Appl. Environ. Microbiol.* **33**, 312–318.
- WINFREY M. R., MARTY D. G., BIANCHI A. J. M. and WARD D. M. (1981) Vertical distribution of sulfate reduction, methane production and bacteria in marine sediments. *Geomicrobiol. J.* **2**, 341–362.
- WOLTEMAATE I., WHITICAR M. J. and SCHOELL M. (1984) Carbon and hydrogen isotopic composition of bacterial methane in a shallow freshwater lake. *Limnol. Oceanogr.* **29**, 985–992.
- WOLTEMAATE I. (1982) Isotopische Untersuchungen zur bakteriellen Gasbildung in einem Süßwassersee. Diplomarbeit, Clausthal Univ., 90 pp.
- ZEIKUS J. G., KERBY R. and KRZYCKI J. A. (1985) Single-carbon chemistry of acetogenic and methanogenic bacteria. *Science* **227**, 1167–1173.
- ZINDER S. H. and MAH R. A. (1979) Isolation and characterization of a thermophilic strain of *Methanosarcina* unable to use H₂O-CO₂ for methanogenesis. *Appl. Environ. Microbiol.* **38**, 996–1008.